Extended Results for Early Detection of Silent Training Failures in Neural Networks

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This file holds the results that did not fit in the paper. It uses the same 885 runs and the same settings ($m = 0.5$, $f = 0.05$, $p = 3$), and every number comes from the saved runs through the scripts in the repository.

1 Networks and runs

The fully connected network has hidden layers of 256, 128, and 64 units. The CNN has three convolutions with 8, 16, and 32 filters, each followed by a ReLU and max pooling, then a dense layer of 64 units and the output layer. The transformer has a width of 64, 4 attention heads, 2 blocks, and feed forward layers of 256 units with a ReLU, so dead units can be counted the same way in all three networks. LHD watches every layer with weights. Units can die in fc1 to fc3 of the fully connected network, in conv1 to conv3 and fc1 of the CNN, and in b0.ff1 and b1.ff1 of the transformer. The CNN is scored on a fixed subset of 4,000 test images and the transformer on 512 fixed windows of 64 characters from the last 10% of the text.

Calibration runs used seeds 0 to 19. Fresh healthy runs and broken runs used seeds from 1000 up, with 10 seeds for each fault and layer, so no scored run touched a range. A dead unit has its incoming weights set to 0 and its bias set to -1. The units are drawn without replacement by a generator seeded with the run's seed plus 7919, and Adam's running averages for them are cleared. Every run trained to the end, and each detector's alarm was found by replaying the saved signals, which gives the same answer as running LHD live.

Runs that share a seed share their starting weights and batch order, and runs that share a fault and layer share the same break, so single runs are not independent. The tests in Section 9 therefore average the gap within each seed and within each fault condition and apply a one sided Wilcoxon signed rank test to those averages. The interval for the mean gap comes from a bootstrap with 10,000 draws that resamples whole seeds, and rates come with 95% Wilson score intervals.

2 Detection over time

Figure E1 shows the share of broken runs each detector had flagged by each epoch. By the last epoch, LHD had flagged 84% of broken runs in the fully connected network, 100% in the transformer, and 78% in the CNN. The accuracy curve had flagged 26%, 48%, and 29%.

3 Every detector

Table E1 lists every detector held to three epochs in a row, with a broken run counted only when the detector named the planted layer. Requiring three epochs instead of one changed little. False alarms did not change,

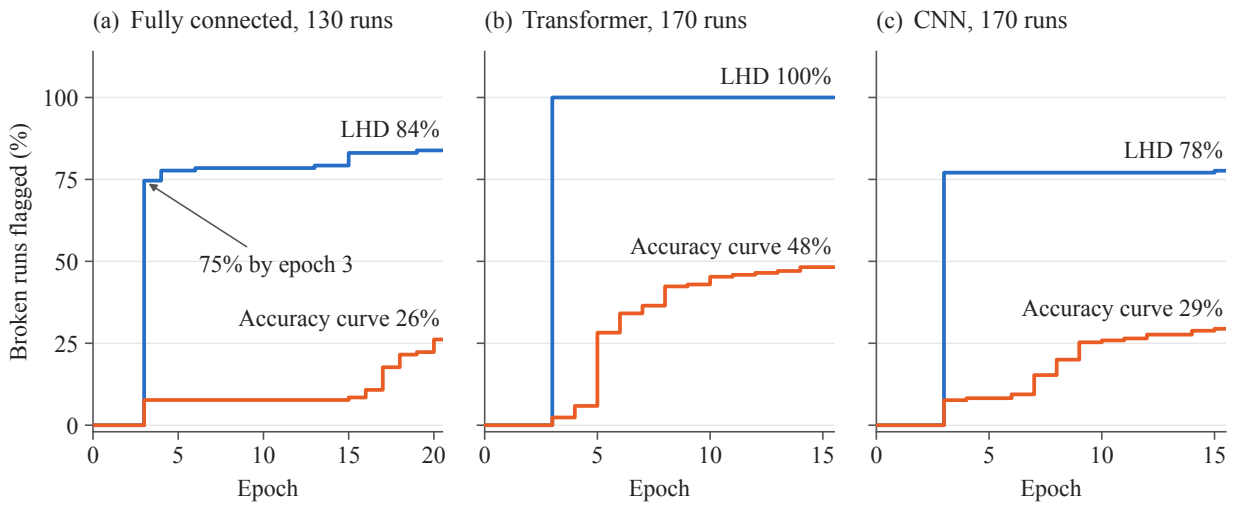


Figure E1: Share of broken runs flagged by each epoch, for LHD and the accuracy curve, in each network.

and dead unit catches fell by at most 2, since a layer's dead unit fraction rarely drops back once it crosses a threshold. Counting only the planted layer mattered more. Counted at any layer, the 50% rule caught more runs with 10% dead units (26) than with 25% (25), because it fired on units that died on their own in other layers. At the planted layer it caught 8 and 10.

Two other curve checks did no better than the accuracy curve in the paper. Requiring only one epoch below the healthy range, the accuracy curve caught 291 of 470 broken runs but flagged 44 of 150 healthy runs. The loss curve, held to three epochs above its range, caught 143 of 470 and flagged 9 of 150.

Table E1: Each detector held to three epochs in a row, with a broken run counted only when the detector named the planted layer. Bold marks the best value in a column when no other detector matches it.

	Caught at the planted layer		
Detector	Frozen layers	Dead units	Healthy runs flagged
LHD	97%	**78%**	3 of 150
Zero update check	**100%**	0%	0 of 150
25% dead rule	0%	71%	51 of 150
SageMaker Debugger (50%)	0%	40%	22 of 150
DeepDiagnosis (70%)	0%	4%	0 of 150
wandb-callbacks (80%)	0%	0%	0 of 150
Any dead unit (wandb-callbacks verbose)	2%	74%	100 of 150

4 Each layer and how many units must die

Figure E2 shows, for every layer and fault, the share of broken runs in which LHD caught the fault and named both the planted layer and the right cause. With 10%, 25%, and 50% of a layer's units dead, LHD caught 57%, 88%, and 90% of runs, against 7%, 9%, and 29% for the accuracy curve. It missed some dead unit runs in the CNN's conv3 at 10% (1 of 10 caught), the CNN's fc1 at 10% (0 of 10 caught), the CNN's fc1 at 25% (0 of 10 caught), the CNN's fc1 at 50% (1 of 10 caught), the fully connected network's fc2 at 10% (0 of 10 caught), the fully connected network's fc3 at 10% (0 of 10 caught), and the fully connected

network's fc3 at 25% (9 of 10 caught). In each of these layers, healthy runs already lose a share of units on their own, so the planted fault stayed inside the healthy range. The CNN's dense layer is the clearest case and shows up as the pale row of the figure.

LHD named a layer other than the planted one in 8 caught runs, namely 50% dead units in fc1 in the CNN reported as fc2 (1), a frozen conv2 in the CNN reported as conv1 (1), a frozen conv3 in the CNN reported as conv2 (1), a frozen fc1 in the CNN reported as conv1 (1), a frozen fc1 in the CNN reported as conv2 (1), and a frozen fc1 in the CNN reported as conv3 (3). In each of them, the layer LHD named had its own signal out of range, so LHD found a real effect of the fault in a different layer. It found the right layer but named the wrong cause in 5 runs, all a frozen b1.qkv in the transformer called a vanishing gradient (5). A frozen attention projection also lowers the gradient that reaches it, so its pattern matches a vanishing gradient.

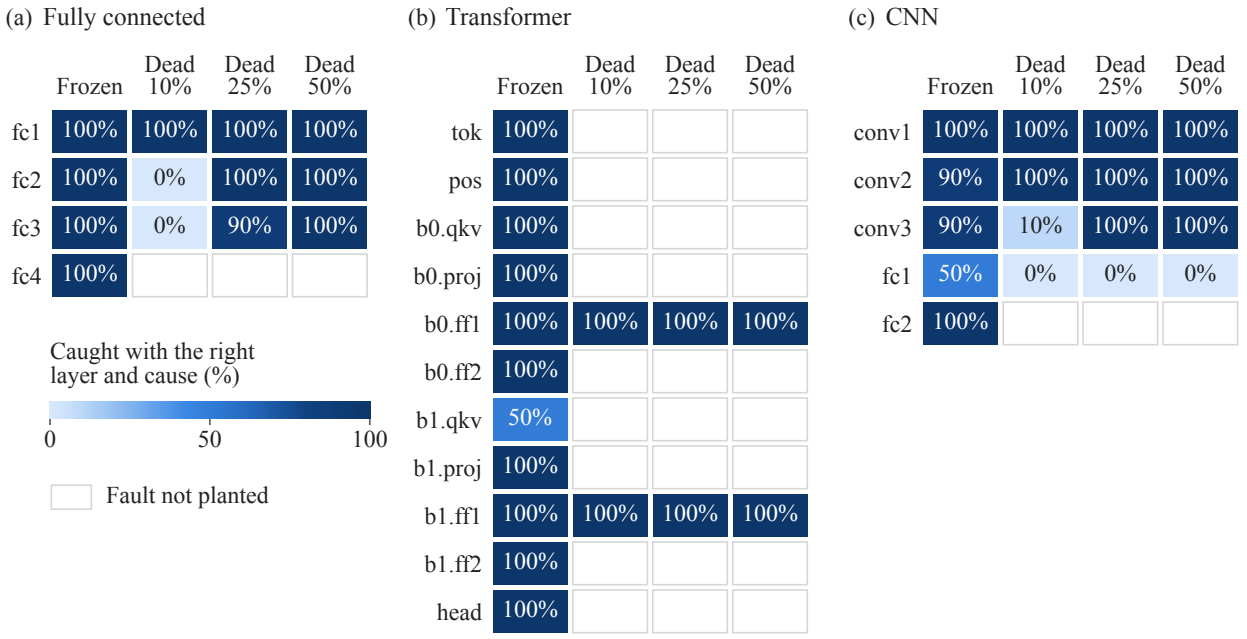


Figure E2: Share of broken runs in which LHD caught the fault and named both the planted layer and the right cause, for every layer and fault. Each cell is 10 broken runs. Blank cells are faults that were not planted, since dead units only go into layers followed by a ReLU.

5 What each signal adds

Table E2 replays the same runs with each signal removed or used alone. Without the dead unit fraction, LHD caught 43% of broken runs instead of 87%, so that signal carries most of the detection. Without the weight change, it named the right layer and cause in 45% of broken runs instead of 85%, since the weight change is what separates a frozen layer from other faults. Removing the weight norm or the gradient norm changed little, because none of the planted faults was designed to move them.

6 Margin and calibration size

These checks replay the same saved runs and were run after the main results. The settings were fixed before them and were not picked from them. With thresholds right at the edge of the healthy range ($m = 0$), LHD

Table E2: What happens when signals are removed or used alone, pooled over the three networks. Changes are in percentage points against all four signals.

Signals used	Broken runs caught		Right layer and cause	
	Rate	Change	Rate	Change
All four	87%		85%	
Without weight change	70%	−18	45%	−40
Without weight norm	87%	same	85%	same
Without grad norm	87%	same	86%	+1
Without dead unit fraction	43%	−45	41%	−44
Only weight change	43%	−45	43%	−42
Only weight norm	10%	−77	0%	−85
Only grad norm	25%	−63	0%	−85
Only dead unit fraction	52%	−36	45%	−40

flagged 71% of healthy runs and caught 98% of broken runs. With the margin of 0.5, it flagged 2% and caught 87%. The range also needs enough calibration runs. Table E3 gives the false alarm rate of the worst network when the range is built from fewer runs, averaged over random draws of those runs.

Table E3: False alarms when the healthy range is built from fewer calibration runs. Each row shows the network with the highest mean false alarm rate at that size.

Calibration runs	Healthy runs flagged	Network
2	22%	CNN
3	15%	CNN
5	11%	fully connected network
10	5%	fully connected network
15	4%	CNN
20	4%	CNN

7 Cost

Monitoring added 11.6% to the training time of the fully connected network, 5.9% for the transformer, and 2.2% for the CNN, measured on fresh healthy runs on one CPU thread. Most of that time goes to the gradient norm, read after every batch, and to the probe batch, run once per epoch, so the overhead is largest for the network with the cheapest epochs. Stopping each broken run at LHD's alarm would have saved 70% of its training on average. All runs together took 107 hours of CPU time on one laptop.

8 Faults beyond frozen layers and dead units

A third set of runs tested faults the rules were not built around (Table E4). It was planned with the rest and trained after the main results were in. Two of these faults are the main ones started at epoch 6 instead of epoch 1, a frozen layer and half of a layer's units killed. A gradient cut detaches the input of one layer from the first epoch, so that layer and the ones after it keep learning while no gradient reaches the layers before it. A learning rate jump multiplies Adam's learning rate by 10 or 100 at the start of epoch 6. The

cosine schedule keeps shrinking the rate from the raised value, so in the 15 epoch networks, epoch 6 trains at 0.0075 or 0.075 instead of 0.00075.

LHD caught 100 of 100 late frozen layers and named the layer and cause in 100, while the accuracy curve caught 20. It caught 40 of 45 runs with half a layer's units killed late and named the layer and cause in 40. It caught 30 of 30 gradient cuts, named a layer before the cut in 30, and called the cause a vanishing gradient, which is the only test of that rule. LHD also fired on every learning rate jump, but the causes it named were mixed. For a tenfold jump it named an unrecognized pattern in 9 runs and dead units in 6 runs, and for a hundredfold jump it named dead units in 5 runs, runaway weights in 5 runs, and a vanishing gradient in 5 runs. No run produced a value that was not a finite number, so the divergence rule never fired.

The zero update check caught every late frozen layer. Held to three epochs it fired in the same epoch as LHD in 100 of 100 runs, and when one epoch is enough it fires at epoch 6. It also caught 30 of 30 gradient cuts, since the layers before a cut never move. The 50% rule caught 37 of 45 late half dead runs at the planted layer, while flagging 22 of 150 healthy runs.

Table E4: Faults beyond the main study, pooled over the three networks. Every detector is held to three epochs in a row. The zero update check and the 50% dead rule count a run only when they name the planted layer, or a layer before a gradient cut, and any alarm counts for a learning rate jump, which has no planted layer. Bold marks the best value in a row when no other detector matches it.

Fault	Runs	LHD	Accuracy curve	Zero update check	50% dead rule
Frozen layer at epoch 6	100	100%	20%	100%	0%
50% dead units at epoch 6	45	**89%**	16%	0%	82%
Gradient cut	30	100%	100%	100%	0%
Learning rate $\times 10$ at epoch 6	15	**100%**	33%	0%	0%
Learning rate $\times 100$ at epoch 6	15	100%	100%	67%	60%
Healthy runs flagged	150	3	7	**0**	22

9 Statistics for each seed and condition

Averaged within each seed, all 30 seed groups favored LHD ($p = 9.3 \times 10^{-10}$), which is the most extreme result a one sided Wilcoxon test can give with that many groups. Averaged within each fault condition, 41 of 47 conditions favored LHD ($p < 10^{-10}$). Of the others, neither detector fired in 2, both fired at the same epoch in 1, and the accuracy curve was earlier in 3, all in layers where LHD missed most of the planted dead units. Counting a run the accuracy curve never caught as one epoch past its end, the mean gap over all broken runs was 9.1 epochs (95% interval 8.4 to 9.7 from the bootstrap over seeds). That figure leans on the runs the accuracy curve missed, so the gap on runs both caught, reported in the paper, is the fairer measure.